(5) (i) $(AB)^{-1} = B^{-1}A^{-1}$. (ii) $(A^{-1})^{-1} = A$

(6) If A is a non-singular square matrix of order n, then

(i) $\left(\text{adj } A\right)^{-1} = \text{adj}\left(A^{-1}\right) = \frac{1}{|A|}A$ (ii) $|\text{adj } A| = |A|^{n-1}$

(iii) adj $\left(\text{adj } A\right) = |A|^{n-2} A$ (iv) $\text{adj}(\lambda A) = \lambda^{n-1}\text{adj}(A)$, λ is a nonzero scalar

(v) $|\text{adj}(\text{adj } A)| = |A|^{(n-1)^2}$ (vi) $(\text{adj } A)^T = \text{adj}\left(A^T\right)$

(vii) $\text{adj}(AB) = (\text{adj } B)(\text{adj } A)$

(7) (i) $A^{-1} = \pm\frac{1}{\sqrt{|\text{adj } A|}}\text{adj } A$. (ii) $A = \pm\frac{1}{\sqrt{|\text{adj } A|}}\text{adj } \left(\text{adj } A\right)$.

(8) (i) A matrix A is orthogonal if $AA^T = A^T A = I$

(ii) A matrix A is orthogonal if and only if A is non-singular and $A^{-1} = A^T$

(8) Methods to solve the system of linear equations $AX = B$

(i) By matrix inversion method $X = A^{-1}B, |A| \neq 0$

(ii) By Cramer's rule $x = \frac{\Delta_1}{\Delta}, y = \frac{\Delta_2}{\Delta}, z = \frac{\Delta_3}{\Delta}, \Delta \neq 0$.

(iii) By Gaussian elimination method

(9) (i) If $\rho(A) = \rho([A \mid B]) =$ number of unknowns, then the system has unique solution.

(ii) If $\rho(A) = \rho([A \mid B]) <$ number of unknowns, then the system has infinitely many solutions.

(iii) If $\rho(A) \neq \rho([A \mid B])$ then the system is inconsistent and has no solution.

(10) The homogenous system of linear equations $AX = 0$

(i) has the trivial solution, if $|A| \neq 0$.

(ii) has a non trivial solution, if $|A| = 0$.

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